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Part One: Ewald Method and Unified Framework

Good morning Professor Wang, today I will present my work on electrostatic interactions, the development of electrostatic interactions in the post-Ewald era and their treatment for machine learning force fields. 11:10/

Let me begin with the fundamental problem. In molecular simulations, to study bulk properties without surface effects, we impose periodic boundary conditions. The simulation box is infinitely replicated, transforming the electrostatic energy from a simple sum over charge pairs into a sum over all periodic images. This infinite series is conditionally convergent, like the alternating harmonic series shown here. Depending on the summation order, you get different values. This conditional convergence is the root of all complexity in periodic electrostatics.

The gold standard solution is Ewald summation. Ewald splits the Coulomb potential into two parts: the short-range $\text{erfc}(r)/r$ and the long-range $\text{erf}(r)/r$. The latter is a convolution of a Gaussian with $1/r$, which is why you see this physical picture in textbooks. Correspondingly, the total energy splits into three parts: a real-space sum, a reciprocal-space sum, and an infinite boundary term. Even today, Ewald summation remains the gold standard.



However, the Ewald method has its pitfalls and misconceptions. First, many find the reciprocal-space sum unintuitive. But if you have a solid background in solid-state physics, the lattice picture I just described should clarify it. Second, Ewald is inherently tied to PBC, which can introduce artifacts and cannot handle isolated systems. Third, it scales as $O(N^2)$, which is computationally expensive. Fourth, it contains an artificially adjustable parameter α .

These limitations drove the field in two directions: abandoning Ewald's concept entirely to develop purely real-space electrostatics, or building upon Ewald's concept to improve computational efficiency.

For the first direction, we must ask: can these purely real-space interactions correctly reproduce the physical properties governed by electrostatics? More importantly, can we determine this directly from their functional forms without running simulations? This motivated our unified framework, connecting functional forms to physical properties.

Before examining these new interactions, we need a

benchmark. The complete electrostatic interaction serves as our ideal reference. But it's conditionally convergent. If you artificially redivide the same lattice into different unit cells, the potential experienced by a charge should theoretically be the same. But it's not. The difference decays as $1/L$, where L is the box size. So while conceptually clear, the complete interaction cannot be used in practice. It exists only as an ideal.

Traditional Ewald's pairwise form omits the infinite boundary term. This corresponds to wrapping the bulk in tin foil to shield boxes at infinity. The boundary term itself takes the form of a three-dimensional limit that depends on how we take that limit. Through Stokes theorem, we convert it into a surface integral combined with a volume limit. Different limiting shapes correspond to different tiling rules. A cube tiles as $1 \times 1 \times 1$, $2 \times 2 \times 2$, $n \times n \times n$. A slab tiles as $n \times \infty \times \infty$. Other shapes have their own elementary function forms. This correspondence between limit paths and tiling rules will be crucial for our physical derivations.

Now, the purely real-space interactions developed in recent decades fall into two categories. First, the angular-averaged Ewald interaction, truncated at r_m , which is the radius of a volume-equivalent sphere. When first proposed, it was called the Yukawa potential. Only in 2022 was it proven that these approximations and complex steps to be the angular average of Ewald. Actually, starting from the summation-order-unspecified form, we can prove this rigorously in one step.

multi- l
Second, the zero-multipole interactions. The idea is to

select charges within a cutoff r_c to make multipole moments up to order l . Zero-order moment gives electroneutrality, also known as Wolf summation. Zero-dipole makes the dipole moment zero. To avoid actually selecting charges every step, Fukuda et al derived functional forms through series expansions. equation in their derivation isn't generally valid, but I've proven it holds for $l \leq 2$, up to l is less than or equal to l .

Part Two: Three Property-Form Links

Now let me present our three property-form links. The first concerns dielectric response. In undergraduate electromagnetics, when an external field is applied to a dielectric, surface charges generate a depolarizing field. The field actually experienced in the bulk is the superposition. The depolarizing field depends on geometry. For a cube, it's $M_{ind}/(3V\epsilon_0)$. For a slab, it's $M_{ind}/(V\epsilon_0)$. But in PBC simulations, the system is infinite with no shape. So, how do we measure the depolarizing field? Through the mathematical form of the boundary term.

When using the cube boundary term, the depolarizing field matches the macroscopic cube. With the plate boundary term, it matches the slab. With no boundary term, there's no depolarizing field at all. Through linear response theory, we proved that the limit path of the boundary term, reflecting the box's tiling rule, is physically equivalent to the macroscopic system's geometric shape.

From this, we vary the ensemble average of the induced

dipole moment. The change in dipole moment is M ind, and the change in field is the applied field. Substituting their relationship gives the dipole fluctuation theorem for different interactions. Some formulas existed before, like tf and cube. De Leeuw and Smith derived them in 1981, using the reaction field picture, assuming an outer dielectric layer with different permittivities. But Frenkel pointed out this is inconsistent. There is no matter beyond infinity. Worse, deriving the plate case required the outer layer's permittivity to be zero, which is impossible; it should at least be one. So for the first time, we've established the dielectric response theory for periodic systems from a unified and correct physical picture.

What about other interactions? They're radially truncated, so each central charge's local environment is spherical, like a cube. For zero-order moment interaction, the local environment has net dipole, generating a depolarizing field consistent with the cube boundary term. For zero-dipole and above, no dipole rearrangement occurs, consistent with tin foil. So our first property-form link is: the limit path of the boundary term represents the box's tiling rule under PBC, equivalent to the macroscopic geometry, and this determines the dielectric response.

The second property is charge-charge long-range correlation. Through Fourier transform of the O-Z equation with the Debye-Hückel approximation, we obtain the correct small-wavevector limit of the long-range correlation function. This limit physically reflects whether a conductor can completely screen an external

field, meaning the depolarizing field exactly cancels the applied field.

For any electrostatic interaction, this limit relates to the Fourier transform of the interaction's unit function. For Ewald, the unit function is the Coulomb potential, with Fourier transform $4\pi/k^2$. Including the reduction factor $1/(4\pi\epsilon_0)$ gives the correct limit.

For angular-averaged Ewald, the Fourier transform doesn't decay as $1/k^2$. But both r_m and k scale with L , so while the limit is incorrect, it doesn't completely diverge. The figure shows simulation results as solid lines and our analytical formula as dashed lines, showing excellent agreement without discrepancy.

For zero-multipole interactions, the cutoff r_c is fixed. So at small wavevectors, $S(k)$ completely diverges. Again, analytics match simulations. These interactions cannot achieve complete screening in simulations.

We extended this to systems with electronic degrees of freedom. For molten salt, the colored lines are simulations, dashed lines our formula, showing strict agreement without discrepancy. So our second property-form link is the decay rate of the unit function's Fourier transform as $k \rightarrow 0$, determines whether the charge-charge long-range correlation's small-wavevector limit is correct, reflecting whether the conductor can completely screen the external field.

The third property is pressure via the virial theorem. In statistical mechanics, pressure comes from differentiating

the configurational integral, giving ideal gas and virial terms. But in periodic systems, potential energy U is an explicit function of volume, so we need an extra volume-dependent term.

Starting from the complete electrostatic interaction, we obtain a simple relation: the volume derivative becomes direct division. This works because all length quantities in the complete interaction are homogeneous in L . Ewald, being a rearrangement, also satisfies this. Even the $1/r$ interaction, which subtracts the boundary term, satisfies it because the boundary term is homogeneous of the same order. Angular-averaged Ewald, being homogeneous in L , also satisfies it without deviation.

But zero-multipole interactions don't satisfy it at all. Take zero-dipole, which has the same form as angular-averaged Ewald. Its cutoff r_c isn't a function of L , breaking homogeneity. Professor Wang Han calculated pressure using zero-dipole and found incorrect results. As r_c increased, pressure converged toward the correct value. Only when r_c equals r_m is the pressure correct. Smaller or larger, it's wrong.

It also explains the pressure correction in one-component plasma. It's non-electroneutral, requiring correction. Previously thought to need a neutralizing screen background with enr interactions. We showed this background isn't physical. It's merely the part containing α in the difference between pairwise-form Ewald and traditional Ewald for non-neutral systems. So our final property-form link is: homogeneity determines the virial theorem form in periodic systems, reflecting

whether pressure calculations are correct.

Part Three: Systematic Error-Free PME

Now let's move to the second direction: building upon Ewald to improve efficiency. This produced particle mesh methods like PME, SPME, and PPPM. But theoretical limitations and technical bottlenecks, especially FFT's global communication, restrict larger-scale applications. D. E. Shaw's Anton still uses PME; Gromacs has begun transitioning to FMM. Nevertheless, analyzing and eliminating PME errors remains essential without compromise.

First, let me briefly introduce PME. It reduces Ewald's reciprocal-space complexity from $O(N^2)$ to $O(N \log N)$. PME discretizes the structure factor by interpolating charges onto a grid, then performing FFT. The reciprocal-space energy takes this form.

PME has three systematic errors. The first is truncation error. Exact Ewald sums wavevectors to infinity. In PME, the summation range couples with grid size. The truncated interaction deviates significantly. For a 100 Å cube, a 16^3 grid gives huge errors. Convergence only occurs at 128^3 . This is why simulation software requires grid spacing under 1 Å.

Our elimination method decouples them. We extend the wavevector sum to a convergent range, then construct a new pairwise interaction grid through one FFT and one inverse FFT. This strictly eliminates truncation error, yielding non-trunc PME. The table shows original PME

needs 128^3 for convergence, while non-trunc PME converges correctly at any grid size without discrepancy. In simulations, the radial distribution function from non-trunc PME far outperforms original PME at the same grid size.

The second error is the discretized charge artifact error, or spurious self-energy. After interpolating point charges onto the grid, interactions arise among these discretized charges that don't exist in the original system. This causes wave-like oscillations in PME's pairwise interaction relative to Ewald. Many have tried eliminating it, but the main difficulty is that each point charge's artifact assembly is necessarily non-neutral. Traditional Ewald cannot handle non-neutral systems.

My advisor solved this in 2014. The pairwise form of Ewald accurately handles non-neutral systems. I adapted it to pairwise PME. We compute each point charge's reciprocal-space energy and subtract it from the total. After eliminating artifact error, oscillations disappear completely. But a discrepancy from exact results remains. The RDF shows further improvement over merely eliminating truncation error. This is for linear interpolation. Higher-order interpolation, especially in SPME, is more complex, but the elimination principle remains unchanged.

The third is fitting error. Shifting perspective from discretized charges back to original charges, PME's long-range interaction is reconstructed by fitting from the interaction grid. This should directly benchmark against true Ewald. It lacks radial symmetry, unsurprising since

~~Force itself isn't radially symmetric.~~ But it also lacks translational invariance, a consequence entirely attributable to grid discretization. The fitting parameters depend separately on the displacement vector's starting and ending points. The discrepancy indicates that Coulomb potential splitting is no longer self-consistent, its integrity compromised.

Our elimination reasoning is simple: in reality, interactions are radially symmetric and translationally invariant. So we perform translational and rotational averaging on the discrepancy. This averaged quantity is necessarily short-ranged, allowing incorporation into the original short-range potential, preserving Coulomb integrity. We eliminate the biased component, the systematic error, while leaving the unbiased component, the fitting noise. Eliminating this noise would require $O(N^2)$ operations, contradicting PME's $O(N \log N)$ efficiency. So unbiased noise is the minimal cost for PME's computational gain.

without systematic errors
After eliminating all systematic errors, we have systematic-error-free PME. For a 100^3 system, original PME needs 128^3 grid for RDF convergence. Systematic-error-free PME achieves convergence with 16^3 or 32^3 . Theoretically, this is over $90\times$ improvement. Practically, with sparser grid and reduced memory, speedup exceeds $100\times$.

To summarize part three: error elimination cost lies mainly in precomputation. During sampling, we add only one step of N subtractions. We've also proven that in particle mesh methods, simultaneously achieving energy

ab initio X-ray data and known static structure factor, we determine a form factor by solving the inverse problem. Back-determining its parameters and inverse transforming to real space yields atom-assignable electron density.

The second difficulty is that under adiabatic dynamics, even without long-range interactions, systematic errors persist. Electronic self-consistency is global. Local equilibrium isn't true global equilibrium. Indirect correlations remain long-ranged. Our proposed solution is that local environment projections shouldn't be completely independent. They can exchange electronic information, analogous to FFT's global communication.

In summary, we identify the spatial nonlocality arising from both long-range correlations and indirect correlations as the primary source of systematic errors in current MLFFs.

Thank you for your attention. I'm happy to take questions.